

PREDICTING HIT SONGS

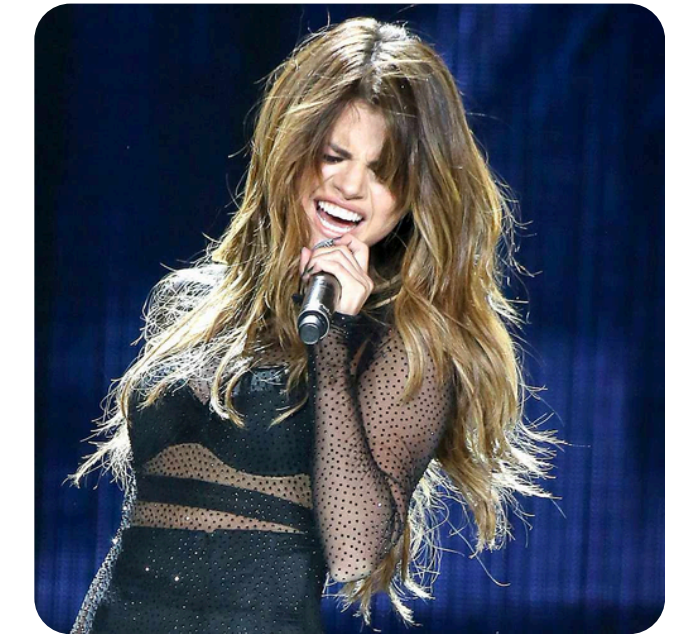
SECTION 1 GROUP 5

Jia Yi Rachel Lee | Maria-Irina Popa |
Thomas Christian Matenco | Roberto Cummings | Enzo Jerez



AGENDA

- 1 Meet the Team
- 2 The Music Industry Dilemma
- 3 The Modelling
- 4 Insights
- 5 Recommendations
- 6 Hands-on Activity



MACHINE LEARNING II

MEET THE TEAM



Jia Yi Rachel Lee



Thomas Christian Matenco



Maria-Irina Popa



Roberto Cummings



Enzo Jerez



MACHINE LEARNING II

Missing Breakout Hits

High-potential songs could be overlooked or underpromoted, leading to lost revenue, streams, and reduced competitive advantage.

Example: *Lady Gaga's Bloody Mary* (delayed success)

**Promoting the Wrong Songs**

Budgets are allocated to tracks that fail to resonate with audiences, eroding marketing efficiency and return on investment.

Example: *Doja Cat's Say So* (not the title track)

**Lack of Scalable Hit-Making Strategy**

An artist's popularity does not always guarantee a hit song.

Example: *Carly Rae Jepsen's Kiss Album* (only Call Me Maybe gained traction)



Smarter
Decisions for
Music
Investments

THE MUSIC INDUSTRY DILEMMA

WHICH SONG WILL BE THE NEXT HIT?

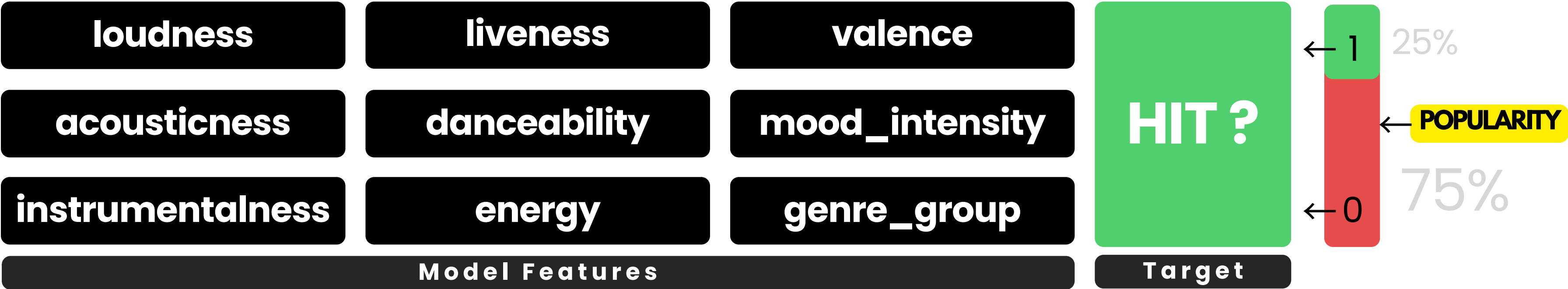
MACHINE LEARNING II

MACHINE LEARNING MODELS

Understanding the Data → Identifying target and features

duration_ms	The track length in Milliseconds	energy	Measure of intensity and activity; high values represent songs that feel fast, loud, and energetic.
explicit	Whether or not the track has any explicit lyrics	instrumentalness	Measure of the likelihood that the track contains no vocals.
danceability	Describes how suitable a track is for dancing (Based on tempo, rhythm stability, beat strength, and overall regularity)	loudness	Overall loudness of the track, measured in decibels.
track_genre	Genre label assigned to the track.	popularity	A score of 0 to 100, reflecting the number of times played

Final selected features



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MACHINE LEARNING MODELS

3 Classification Models were explored: Classification Tree, Random Forest, Boosting

Classification Tree:

- Interpretable baseline
- High variance and overfitting risk



Random Forest:

- Reduces variance through ensembling
- More stable, but limited improvement



♥ XGBoost

- Learns sequentially from errors
- Best overall predictive performance

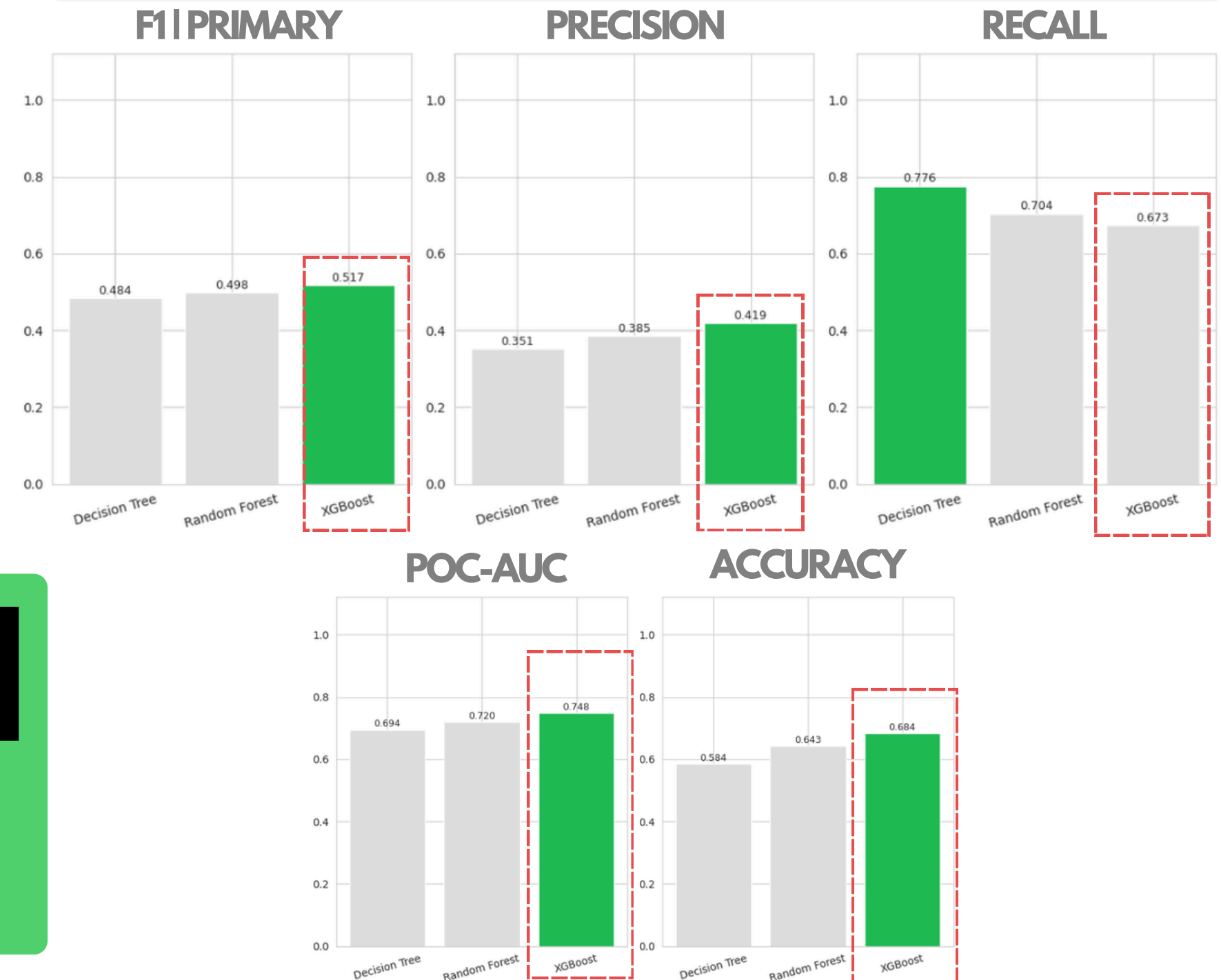


XGBoost

- Best overall across most metrics
- Highest F1, precision, ROC-AUC, and accuracy
- Best trade-off between performance and generalisation
- Recall difference was small, but gains in F1, ROC-AUC, precision, and accuracy were meaningful



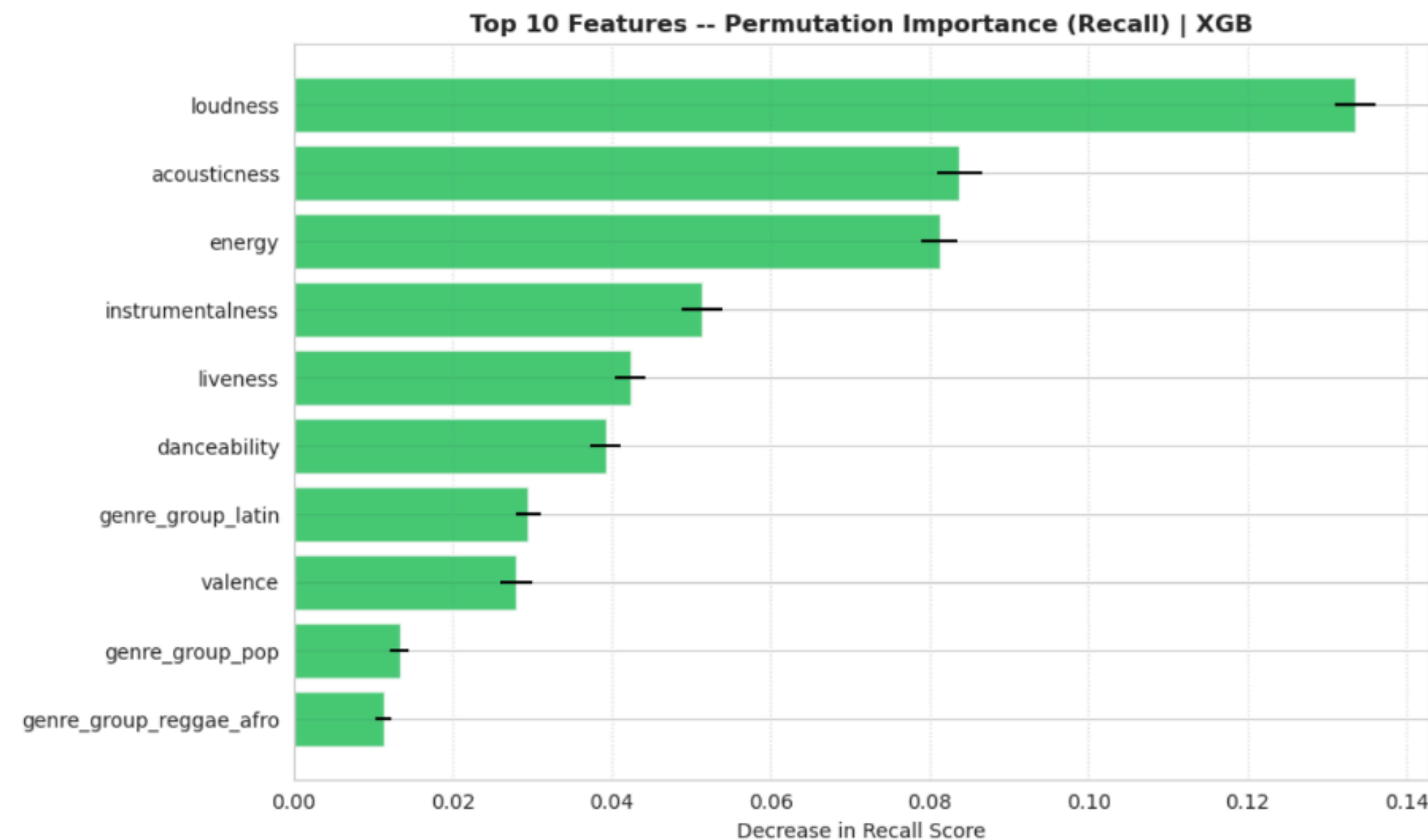
Model Comparison



MACHINE LEARNING II

AUDIO FEATURES ARE STRONGER DRIVERS OF A SONG'S SUCCESS

Permutation Importance:



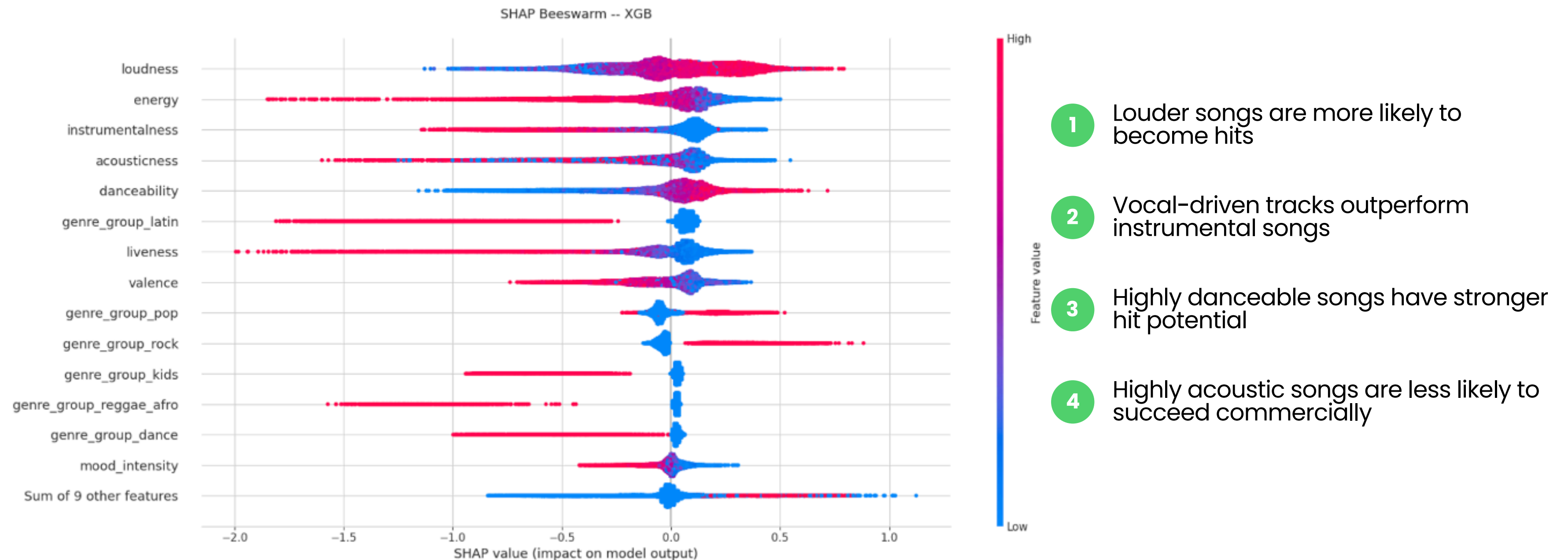
Audio features play a more significant role in determining a song's success than genre

- 1 Loudness
- 2 Acousticness
- 3 Energy

MACHINE LEARNING II

AUDIO FEATURES ARE STRONGER DRIVERS OF A SONG'S SUCCESS

SHAP Analysis:



MACHINE LEARNING II

AUDIO FEATURES ARE STRONGER DRIVERS OF A SONG'S SUCCESS

Using the top 10 features, and combining with insights from SHAP analysis:



Ahora Te Puedes
Marchar



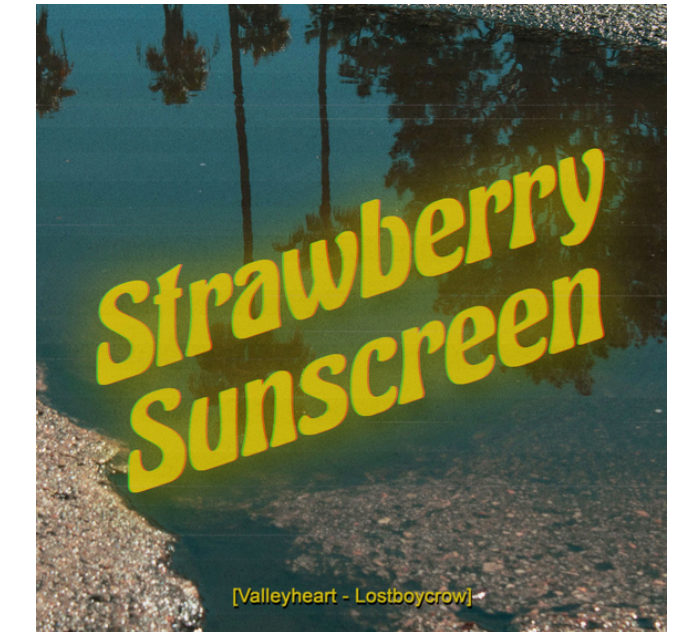
What Makes You
Beautiful



Beatbox



Gentleman



Strawberry
Sunscreen

MACHINE LEARNING II

IDENTIFYING POTENTIAL HITS IS KEY

Simulating the economics of the music industry:

False Positive (FP):	Cost of promoting a non-hit song	€1,000
False Negative (FN):	Opportunity cost of missing a hit song	€5,000
True Positive (TP):	Gains from promoting a hit song	€10,000

Scenario 1

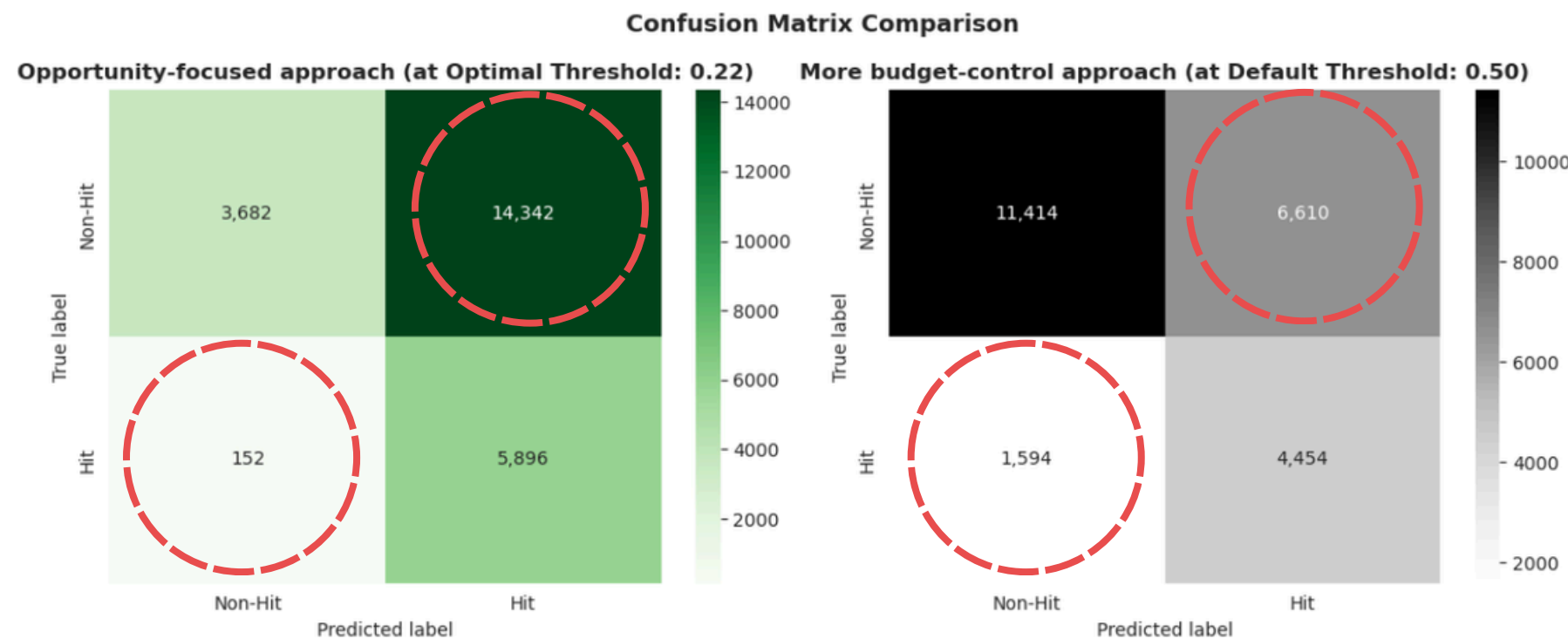
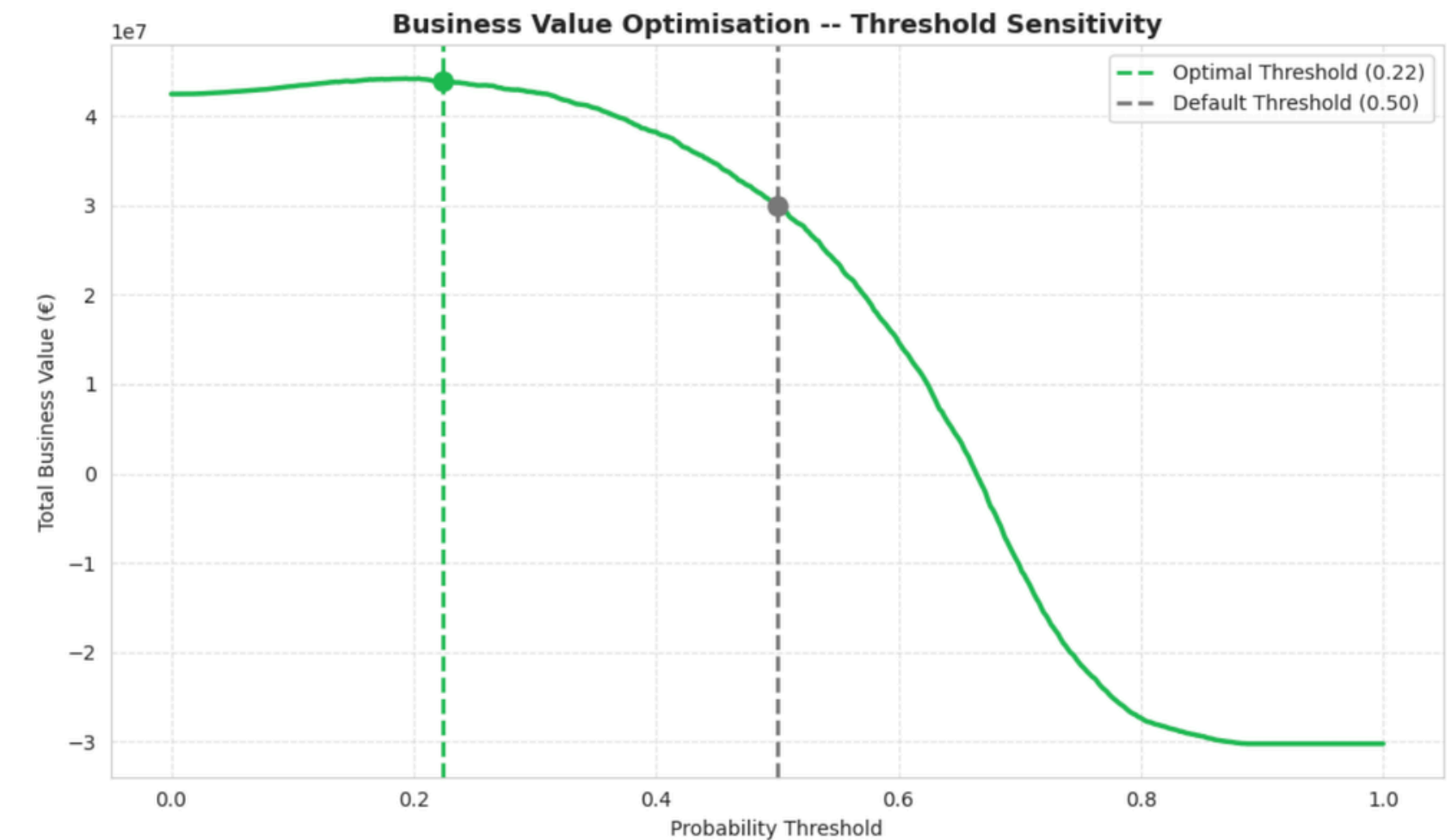
Wasted Cost = 14342 x €1000
Opp Cost = 152 x €5000
Revenue = 5896 x €10000

**Profit
= €43.8M**

Scenario 2

Wasted Cost = 6610 x €1000
Opp Cost = 1594 x €5000
Revenue = 4454 x €10000

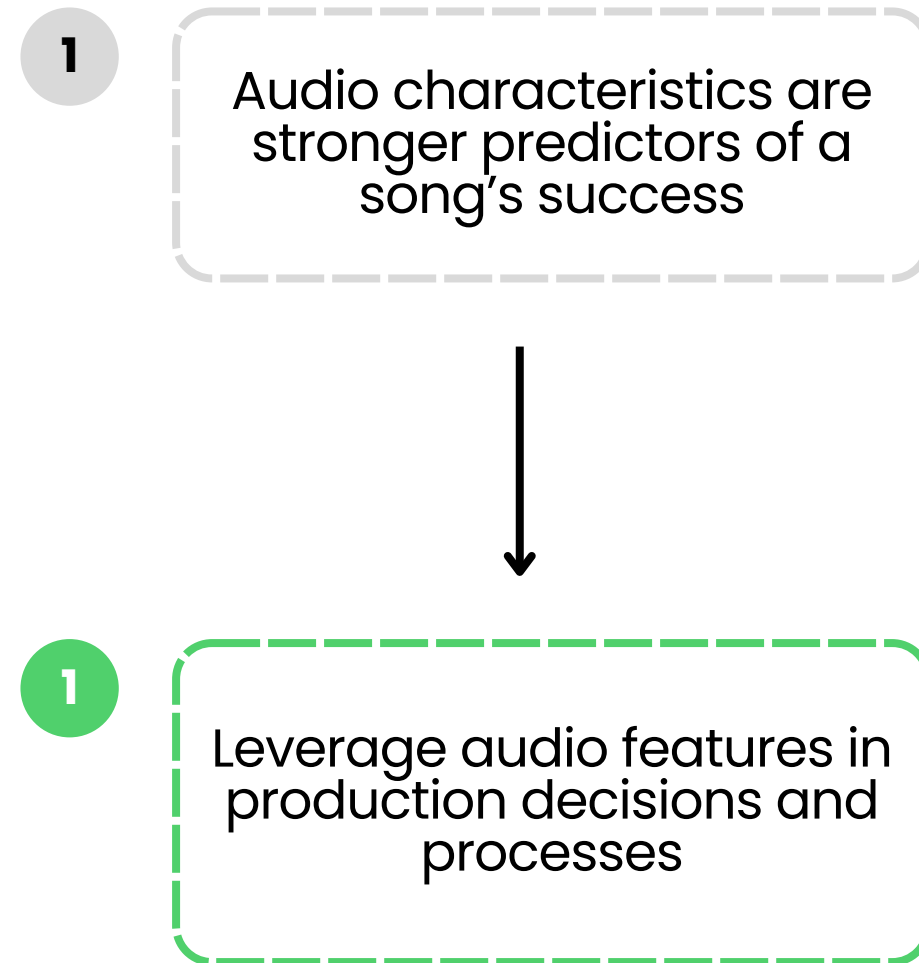
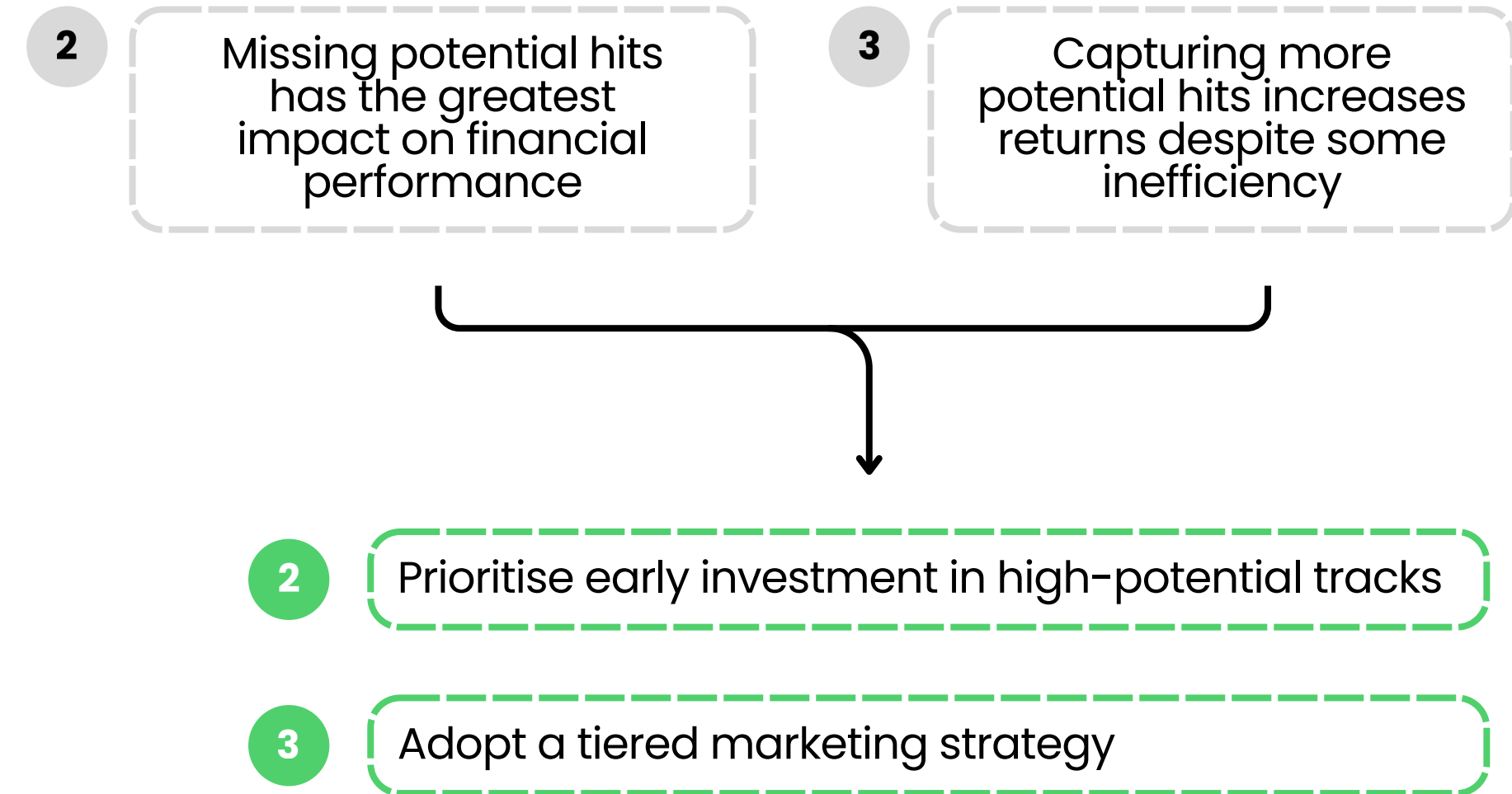
**Profit
= €30.0M**



Capturing more potential hits drives higher financial returns than avoiding wasted promotion

MACHINE LEARNING II

TRANSLATING INSIGHTS INTO DATA DRIVEN STRATEGY FOR THE MUSIC INDUSTRY

Audio Components**Business Implications Considerations**

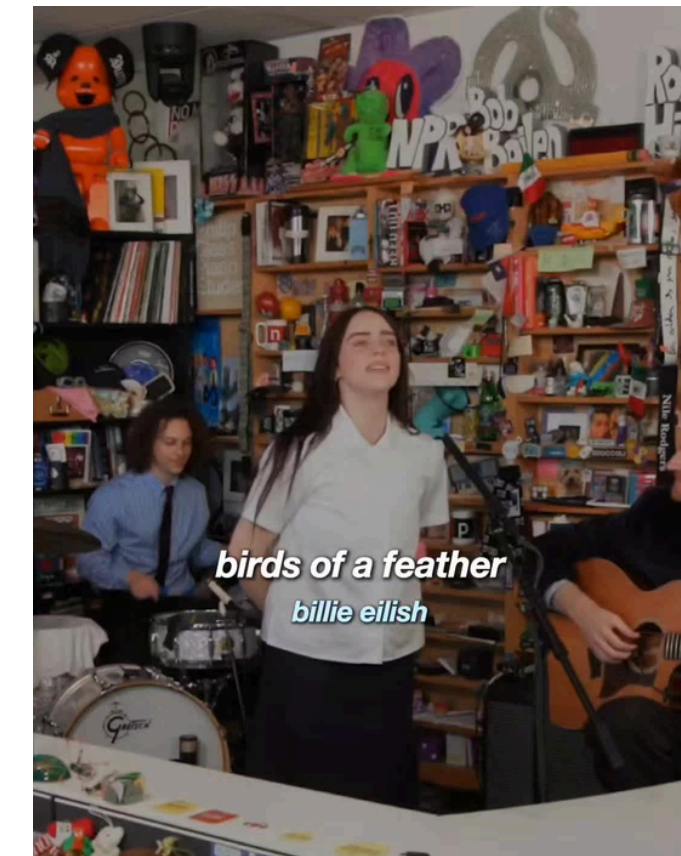
Ultimately, record labels should consider their risk appetite when deciding their investment strategy - achieving a balance between maximising hit discovery and managing marketing budget is key

MACHINE LEARNING II

ACHIEVING HIGHER RETURNS THROUGH SCALING HITS IDENTIFICATION

XG Boosting**Identify songs with
high potential to be
hit songs****Strategic
Investments in
these songs****Marketing &
Promotion****Success?**

**A broader opportunity
focused approach to
identifying hits deliver
higher overall returns
eventually**

**Limitations:**

- Artist influence is not included
- Changing musical preferences by the decades
- Impact of social media and virality is not considered
- Analysis is based on Spotify data only

MACHINE LEARNING II

1. PICK AN ARTIST

2. PICK A SONG

Example:

```
=====
[HIT] Blank Space – Taylor Swift
Genre: pop | Popularity: 85/100
=====
```

Feature	Value	vs Hits	Signal
Loudness	-5.4 dB	top 68.5%	[+]
Acousticness	0.103	top 58.4%	[+]
Energy	0.703	top 43.5%	[-]
Instrumentalness	0.0000	top 100.0%	[+]
Liveness	0.091	top 79.7%	[+]
Danceability	0.760	top 86.3%	[+]
Valence	0.570	top 68.0%	[+]
Mood Intensity	0.401	top 69.6%	[+]
Genre			[+] pop – positive signal

Feature score: 8/9 features align with hit profile

Overall Hit Score: 79.5 / 100
(100 = perfect hit profile across all features)

THANK YOU

